

MILITARY SPECIFICATION

MODIFICATION KIT, AIRCREW PROTECTIVE HELMET KMU-441/P

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of modification kit for an aircrew protective helmet. The modification kit is designated KMU-441/P.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

C-F-206	Felt Sheet: Cloth, Felt, Wool, Pressed
L-P-378	Plastic Sheet and Strip, Thin Gauge, Polyolefin
L-P-386	Plastic Material, Cellular, Urethane (Flexible)
V-T-295	Thread, Nylon
GG-D-226	Depressors, Tongue
KK-L-254	Leather, Sheepskin, Chrome Tanned
CCC-C-432	Cloth, Sheeting, Cotton, (Unbleached, Bleached, and Dyed)
DDD-L-20	Labels: for Clothing, Equipage, and Tentage (General Use)
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-676	Boxes, Setup

MILITARY

MIL-W-5664	Webbing, Textile, Elastic, Cotton
MIL-R-19907	Repair Kit, Glass Reinforced Plastic Laminate
MIL-F-21840	Fastener Tapes, Hook and Pile, Synthetic
MIL-C-40004	Cloth, Knitted, Cotton, Simplex

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-751 Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 Marking for Shipment and Storage
MIL-STD-130 Identification Marking of US Military Property

DRAWINGS

AIR FORCE

64A2217 Pads - Fitting, Flying Helmet
68B270 Pads - Earcup, Flying Helmet
70359 Patterns - Liner - Absorbent, Flying Helmet
70360 Liner Assembly - Absorbent, Flying Helmet
76139 Pad - Chin and Nape, Flying Helmet

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AEROSPACE MATERIALS SPECIFICATIONS (AMS)

AMS 3570 Flexible Polyurethane Foam

(Copies of this specification can be obtained from the Society of Automotive Engineers, Inc.; 485 Lexington Avenue; New York, New York 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries and are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Major components. The major components of the modification kit shall consist of the essential items required to modify ten helmets to include liners and shall consist of the items listed in table I.

3.2 Materials

3.2.1 Plastic foam. The front pads and the back pads shall be made of flexible urethane foam or of flexible polyurethane foam. The flexible urethane foam shall be 1/4 inch thick, shall be white or natural color, and shall

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conform to type II, class 1 of L-P-386. The flexible polyurethane foam shall be 1/4 inch thick, shall be charcoal gray color, and shall conform to AMS 3570 (National Stock No. 9330-00-300-1531).

TABLE I. Major components.

Item No.	Quantities	Nomenclatures	References
1	10	Pad, Helmet Liner, Front	3.4.1
2	10	Pad, Helmet Liner, Back	3.4.2
3	10	Cover, Front Liner Pad	3.4.3
4	10	Cover, Back Liner Pad	3.4.4
5	10	Protective Cover	3.4.5
6	10 sets	Pads, Earcup	3.4.6
7	4	Absorbent Liner, Medium	3.4.7
8	6	Absorbent Liner, Large	3.4.7
9	20	Pad, Chin and Nape, Flying Helmet	3.4.8
10	1	Kit, Helmet Shell Repair	3.4.9

3.2.2 Leather. The leather for the front pad covers and the back pad covers shall be black and shall conform to type A, class 2, subclass b, drum method of application of KK-L-254.

3.2.3 Plastic film. The plastic film for the protective covers shall be 0.002 inch thick and shall conform to type I, class 1, grade A, finish 1 of L-P-378. The plastic film for the plastic bags shall conform to type I, class 1, grade A, finish 2 of L-P-378.

3.2.4 Edge binding. The material for the edge binding of the absorbent liners shall be gray and shall conform to type VI, class 3 of CCC-C-432.

3.2.5 Cotton cloth. The cotton cloth for the absorbent liners shall conform to type II; class 2, 3, or 5 of MIL-C-40004. The cotton cloth for the covers of the chin and nape pads shall conform to type II, class 2 or 3 of MIL-C-40004.

3.2.6 Thread. The thread for sewing the absorbent liners shall be white or gray color and shall conform to type I, size A or B of V-T-295. The thread for sewing the chin and nape pads shall be white and shall conform to type I, size B of V-T-295.

3.2.7 Felt. The felt for the chin and nape pads shall be 1/4 inch thick and shall conform to type I, classification 8R5 of C-F-206.

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3.2.8 Webbing. The webbing for the chin and nape pads shall be 1/2 inch wide and shall conform to class 1 of MIL-W-5664.

3.2.9 Backing of chin and nape pads. The backing of the chin and nape pads shall be Fiberthin, Type C1821, white (manufactured by the US Rubber Co.; Mishawaka, Indiana), or equal.

3.2.10 Polyurethane foam. The polyurethane foam for the earcup pads shall be white, resilient polyurethane foam with a density of 3.5 to 4.0 pounds per cubic foot. The thickness of the polyurethane foam shall be as specified on Drawing 68B270.

3.2.11 Plastic tape. The tape for the protective covers shall be a transparent, plastic, pressure-sensitive adhesive tape.

3.2.12 Hook fastener tape. The hook fastener tape for the earcup pads shall be white, shall be 2 inches wide, and shall conform to type I, class 1 of MIL-F-21840.

3.2.13 Material for pad backing. The material for the pad backing of the earcups shall be as specified on Drawing 68B270.

3.3 Design and construction. The modification kit shall be designed and constructed in accordance with the requirements specified herein and in accordance with the requirements of the drawings referenced herein. In the event of conflict between the requirements of the drawings referenced herein and the requirements of the text of this specification, the requirements of the text of this specification shall apply.

3.3.1 Stitches, seams, and stitchings. Stitches, seams, and stitchings used in the construction of the absorbent liners and the chin and nape pads shall conform to FED-STD-751.

3.4 Details of components. The components of the modification kit shall be assembled as specified in 3.5. Unless otherwise specified herein, the closure of the plastic bags shall not be heat sealed.

3.4.1 Front pads. Each front pad shall be 12 inches long, 10 inches wide, and 1/4 inch thick. The tolerance for the length, the width, and the thickness of each front pad shall be as specified in L-P-386.

3.4.2 Back pads. Each back pad shall be 12 inches long, 12 inches wide, and 1/4 inch thick. The tolerance for the length, the width, and the thickness of each front pad shall be as specified in L-P-386.

3.4.3 Front pad covers. Each front pad cover shall be 14 inches (plus 1/4 or minus 1/8 inch) long and 10 inches (plus 1/4 or minus 1/8 inch) wide.

3.4.4 Back pad covers. Each back pad cover shall be 14 inches (plus 1/4 or minus 1/8 inch) long and 11 inches (plus 1/4 or minus 1/8 inch) wide.

3.4.5 Protective covers. Each protective cover shall conform to figure 1.

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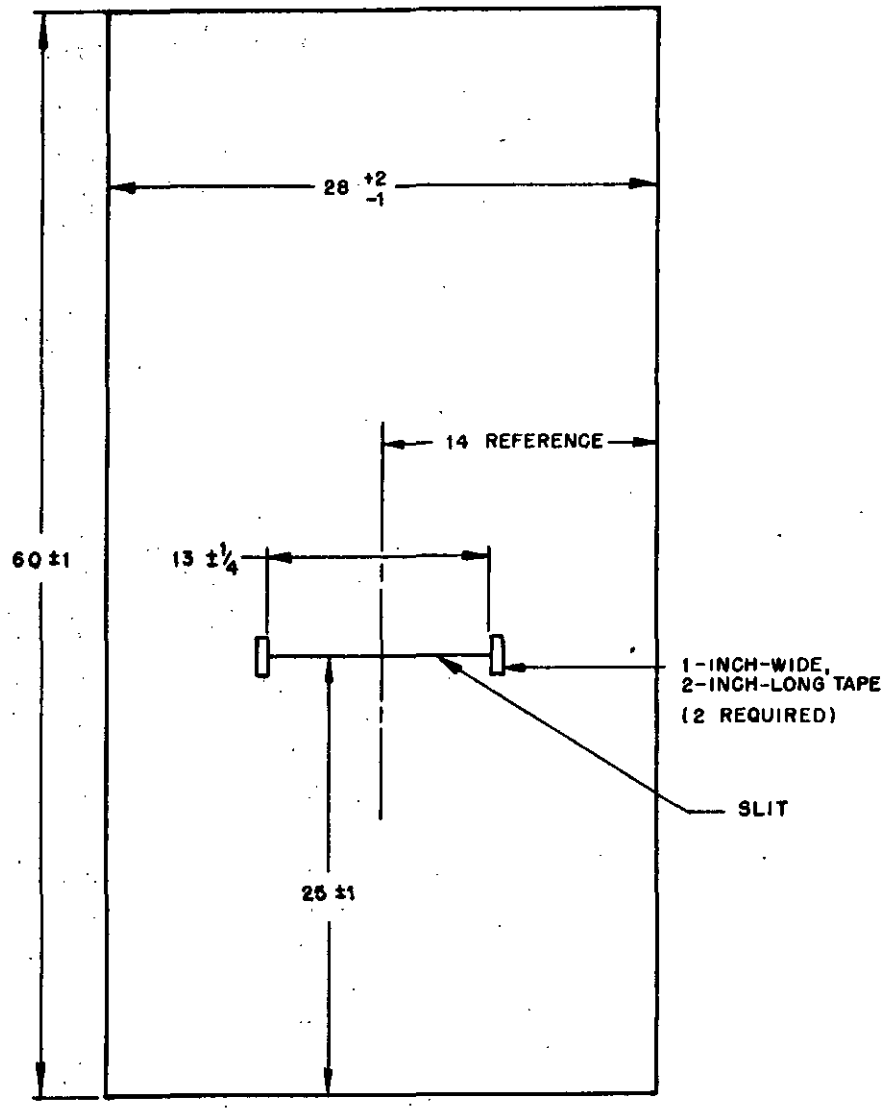


FIGURE 1. Protective cover.

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3.4.6 Earcup pads. The earcup pads shall conform to Drawing 68B270 except that the requirements for packaging shall not apply. Each set of the earcup pads shall consist of the components specified for -4 on Drawing 64A2217 except that the earcup installation tapes will not be required.

3.4.7 Absorbent liners. Each absorbent liner in size medium and each absorbent liner in size large shall conform to Drawings 70359 and 70360.

3.4.8 Chin and nape pads. Each chin and nape pad shall conform to Drawing 76139.

3.4.9 Repair kit. The repair kit shall consist of two spatulas, one tube of resin, and one tube of hardener. The spatulas shall be tongue depressors that conform to GG-D-226. The resin (including the tube) shall conform to the requirements specified under "Paste Resin" in section 3 of MIL-R-19907. The hardener (including the tube) shall conform to the requirements specified under "Paste Hardener" in section 3 of MIL-R-19907. The tube material shall not affect the paste hardener nor be affected by the paste hardener. The tube of the paste resin and the tube of the paste hardener shall conform to the requirements specified under "Performance Requirements for Paste Resin and Hardener" in section 3 of MIL-R-19907.

3.5 Assembly of major components. Unless otherwise specified herein, the closure of the plastic bags in which the components are assembled shall not be heat sealed. The components of the modification kit shall be assembled as follows:

(a) The ten front pads shall be laid flat, stacked ten high, and then placed in a plastic bag.

(b) The ten back pads shall be laid flat, stacked ten high, and then placed in a plastic bag.

(c) The ten front pad covers shall be laid flat, stacked ten high, and then placed in a plastic bag.

(d) The ten back pad covers shall be laid flat, stacked ten high, and placed in a plastic bag.

(e) The ten protective covers shall be individually folded to a size that is approximately 14 inches by 15 inches, stacked ten high, and then placed in a plastic bag.

(f) Each set of the earcups shall be placed in a plastic bag. The closures of the ten plastic bags shall be heat sealed. The ten plastic bags (each containing one set of earcups) shall be placed in a plastic bag.

(g) Each absorbent liner in size medium shall be placed in a plastic bag. The closures of the four plastic bags shall be heat sealed. The four plastic bags (each containing one absorbent liner in size medium) shall be placed in a plastic bag.

(h) Each absorbent liner in size large shall be placed in a plastic bag. The closures of the six plastic bags shall be heat sealed. The six plastic bags (each containing one absorbent liner in size large) shall be placed in a plastic bag.

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(i) The twenty nape and chin pads shall be placed in a plastic bag.

(j) The screw caps of the tube of paste resin and of the tube of paste hardener shall be closed. The tube of paste resin, the tube of paste hardener, and the two spatulas shall be placed in a snug-fitting box that conforms to PPP-B-566, class domestic of PPP-B-636, or PPP-B-676. The box shall be closed as specified in the applicable box specification or appendix thereto.

3.6 Identification of product

3.6.1 Plastic bags. Each plastic bag specified in 3.5 shall be marked for identification or shall contain an identification label. Each identification label or the outside of each plastic bag shall be marked in accordance with MIL-STD-130. The marking shall include the nomenclature, the size, and the quantity of each item within the plastic bag. The nomenclature shall be as listed in table I.

3.6.2 Absorbent liners. The exterior of a side panel of each absorbent liner shall be marked for identification in accordance with type IV, class 2 of DDD-L-20. The item identification for the absorbent liners shall be as listed in table I.

3.6.3 Components of repair kit. The tube that contains the paste resin shall be marked for identification in accordance with the paragraph entitled "Labeling for Paste Resin Tube" in section 5 of MIL-R-19907. The tube that contains the paste hardener shall be marked for identification in accordance with the paragraph entitled "Labeling for Paste Hardener Container" in section 5 of MIL-R-19907. The box that contains the repair kit shall be marked for identification in accordance with the paragraph entitled "Labeling for Paste Resin-Hardener Package" in section 5 of MIL-R-19907.

3.7 Workmanship. The components of the modification kit shall be manufactured in a thoroughly workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsibile for the performance of all the inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Certificates of compliance. If a certificate of compliance is furnished for any requirement specified herein, the Government shall have the right to check test items to determine the validity of the certification.

4.3 Quality conformance inspection

4.3.1 Materials and components. Unless otherwise specified herein, the materials and the components to be used in the construction of the modification kits shall be inspected in accordance with the subsidiary documents referenced herein to the extent applicable. The paste resin and the paste hardener (see

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3.4.9) shall be tested in accordance with the tests described under "Paste Resin and Paste Hardener" and "Paste Resin Hardener Mixture" in section 4 or MIL-R-19907. The spatulas shall be tested and examined as specified in GG-D-226. The unit for the expression of the lot size, the sample size, and the acceptable quality level for the inspection of the materials and the components shall be as specified in the applicable subsidiary documents.

4.3.2 End items

4.3.2.1 Individual inspection. Each assembled modification kit (see 3.1) shall be subjected to the visual examination described in 4.4.1.

4.3.2.2 Sampling inspection. Samples of the modification kits shall be subjected to the dimensional examination described in 4.4.2. The sample unit and the unit for the expression of the lot size for this dimensional examination shall be one assembled modification kit (see 3.1). The inspection level for this dimensional examination shall be S-2 of MIL-STD-105. The acceptable quality level for this dimensional examination shall be 4.0 defects per 100 units (one class).

4.3.3 Packaging. Samples of the shipping containers fully prepared for delivery shall be subjected to the examination of packaging described in 4.4.3. The sample unit and the unit for the expression of the lot size for this examination of packaging shall be one shipping container fully prepared for delivery except that it need not be sealed. The lot size for this examination of packaging shall be the number of shipping containers in the inspection lot. The inspection level for this examination of packaging shall be S-2 of MIL-STD-105. The acceptable quality level for this examination of packaging shall be 2.5 defects per 100 units (one class).

4.4 Inspection methods

4.4.1 Visual examination. The components of the assembled modification kit shall be visually examined to determine conformance to the requirements specified herein with respect to quantity, materials, design, construction, identification marking, and workmanship.

4.4.2 Dimensional examination. Except for the repair kit, the components of the assembled modification kit shall be examined to determine conformance to the dimensions specified herein or on the drawings referenced herein. Each dimension that is not as specified herein or on the drawings referenced herein shall be classified as a defect.

4.4.3 Examination of packaging. The shipping container fully prepared for delivery except that it need not be sealed shall be examined, for the defects listed in table II, to determine conformance to the requirements specified in section 5 of this specification.

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TABLE II. Examination of packaging.

Examine	Defects
Marking (interior and exterior)	Marking omitted; incorrect; illegible; or improper size, location, sequence, or method of application
Materials	Any component missing, damage, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling
Weight	Gross weight exceeds requirements

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The major components of one modification kit that have been assembled as specified in 3.5 shall be packaged in a snug-fitting box (approximately 36 inches by 15 inches by 6 inches). The box shall conform to type CF, class domestic, variety SW, grade 175 of PPP-B-636.

5.1.2 Level C. Modification kits that have been assembled as specified in 3.5 shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Modification kits that have been packaged as specified shall be packed in box that conforms to type CF, class weather-resistant, variety DW, grade V13c of PPP-B-636. Closure of the box shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. Modification kits that have been packaged as specified shall be packed in box that conforms to type CF, class weather-resistant, variety DW, grade 200 of PPP-B-636. Closure of the box shall be as specified in the appendix of PPP-B-636.

5.2.3 Level C. Modification kits that have been packaged as specified shall be packed in a manner that will afford adequate protection, at the lowest transportation rate, against damage during direct shipment from the supply source to the first receiving activity for immediate use. The shipping containers shall conform to the rules of carriers or the regulations applicable to the mode of transportation .

5.3 Marking. In addition to any special marking required by the contract or order and unless otherwise specified herein, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Plastic bags. The plastic bags that contain the components of the modification kits shall be marked as specified in 3.6.1.

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5.3.2 Repair kit. The box that contains the repair kit shall be marked as specified in 3.6.3.

6. NOTES

6.1 Intended use. The modification kit covered by this specification is intended to provide the materials and the components for finishing ten HGU-26/P helmets during base level modification of the helmets.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number, title, and date of this specification

(b) Selection of applicable levels of preservation and packaging and packing (see 5.1 and 5.2).

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Air Force - 11

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